Chapter 4

Market Impact Models

INTRODUCTION

This chapter provides an overview of market impact models with emphasis on the "Almgren & Chriss" and the "I-Star" approaches. The Almgren and Chriss (AC) model, introduced by Robert Almgren and Neil Chriss (1997), is a path-dependent approach that estimates costs for an entire order based on the sequence of trades. This is referred to as a bottom-up approach because the cost for the entire order is determined from the actual sequence of trades.

The I-Star model, introduced by Robert Kissell and Roberto Malamut (1998), is a top-down cost allocation approach. First, we calculate the cost of the entire order, and then allocate to trade periods based on the actual trade schedule (trade trajectory). The preferred I-Star formulation is a power function incorporating imbalance (size), volatility, liquidity, and intraday trading patterns.

Alternative market impact modeling approaches have also appeared in the academic literature. For example, Wagner (1991); Kissell and Glantz (2003); Chan and Lakonishok (1997); Keim and Madhavan (1997); Barra (1997); Bertismas and Lo (1998); Breen, Hodrick, and Korajczyk (2002); Lillo, Farmer, and Mantegna (2003); and Gatheral (2010, 2012).

DEFINITION

Market impact is the change in price caused by a particular trade or order. It is one of the more costly transaction cost components and always causes adverse price movement. Market impact is often the main reason managers lag behind their peers. Market impact costs will occur for two reasons: liquidity needs and urgency demands (temporary), and information content (permanent).

Temporary Impact represents the liquidity and urgency cost component. This is the price premium buyers have to provide the market to attract additional sellers and the price discount sellers need to provide

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